

```
import numpy as np
import pandas as pd
import matplotlib as plt
```

raw_df

	respondent_id	q2_1	q2_2	q2_3	q3_18	q3_19	q3_20	q3_21	q3_22	q3_23	...	Q24_6_TEXT	q7_1	q7_2	q7_3	q7_4	q7_5	category
0	jtn-31kgg8psdBCqYD6	1	0	0	0	0	0	0	0	0	...	NaN	8	8	8	8	8	R
1	jtn-5FQ679JNrH6g7dT	1	0	0	0	0	0	0	0	0	...	NaN	4	1	3	1	4	R
2	jtn-22853lleE3oxpXj	1	0	0	0	0	0	0	0	0	...	NaN	4	4	4	4	4	H
3	jtn-1491R1Uo7f7tb0f	0	1	0	0	0	0	0	0	0	...	NaN	8	8	8	8	8	R
4	jtn-2UaYA52mtFm5pF3	1	0	0	0	0	0	0	0	0	...	NaN	5	5	5	5	6	H
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
3089	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	0	1	0	0	0	0	0	0	0	...	NaN	4	4	4	4	4	...
3090	jtn-vG9F16NqfRwxVLx	0	1	0	0	0	0	0	0	0	...	NaN	6	3	6	8	8	E
3091	jtn-4guyjfQ0PtcYI91	0	1	0	0	0	0	0	0	0	...	NaN	1	1	1	8	1	R
3092	jtn-za370CyvD9fcoZn	1	0	0	0	0	0	0	0	0	...	NaN	6	5	6	5	6	H
3093	jtn-uqgmsyrCPGOMYf8	1	0	0	0	0	0	0	0	0	...	NaN	4	4	4	4	1	R

3094 rows × 71 columns

df_q7_long

	respondent_id	question_base	precode	question_id	description
0	jtn-31kgg8psdBCqYD6	q7_1	8	q7_1_8	Help me discover a new brand, product or service
1	jtn-5FQ679JNrH6g7dT	q7_1	4	q7_1_4	Help me discover a new brand, product or service
2	jtn-22853lleE3oxpXj	q7_1	4	q7_1_4	Help me discover a new brand, product or service
3	jtn-1491R1Uo7f7tb0f	q7_1	8	q7_1_8	Help me discover a new brand, product or service
4	jtn-2UaYA52mtFm5pF3	q7_1	5	q7_1_5	Help me discover a new brand, product or service
...	...	...	...	...	...
15465	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	q7_5	4	q7_5_4	To help me find the best brand/product relevan...
15466	jtn-vG9F16NqfRwxVLx	q7_5	8	q7_5_8	To help me find the best brand/product relevan...
15467	jtn-4guyjfQ0PtcYI91	q7_5	1	q7_5_1	To help me find the best brand/product relevan...
15468	jtn-za370CyvD9fcoZn	q7_5	6	q7_5_6	To help me find the best brand/product relevan...
15469	jtn-uqgmsyrCPGOMYf8	q7_5	1	q7_5_1	To help me find the best brand/product relevan...

15470 rows × 5 columns

```
[f'q4_{i}' for i in range(1, 5)]
```

```
['q4_1', 'q4_2', 'q4_3', 'q4_4']
```

```
cross_tab = df_q7_long.groupby(['question_base', 'question_id']).size().reset_index(name='count')
print(cross_tab)
```

	question_base	question_id	count
0	q7_1	q7_1_1	639
1	q7_1	q7_1_2	283

```
2      q7_1      q7_1_3      372
3      q7_1      q7_1_4      539
4      q7_1      q7_1_5      487
5      q7_1      q7_1_6      315
6      q7_1      q7_1_7      121
7      q7_1      q7_1_8      338
8      q7_2      q7_2_1      514
9      q7_2      q7_2_2      211
10     q7_2      q7_2_3      387
11     q7_2      q7_2_4      641
12     q7_2      q7_2_5      494
13     q7_2      q7_2_6      270
14     q7_2      q7_2_7      157
15     q7_2      q7_2_8      420
16     q7_3      q7_3_1      490
17     q7_3      q7_3_2      181
18     q7_3      q7_3_3      323
19     q7_3      q7_3_4      743
20     q7_3      q7_3_5      446
21     q7_3      q7_3_6      265
22     q7_3      q7_3_7      160
23     q7_3      q7_3_8      486
24     q7_4      q7_4_1      517
25     q7_4      q7_4_2      158
26     q7_4      q7_4_3      248
27     q7_4      q7_4_4      1334
28     q7_4      q7_4_5      245
29     q7_4      q7_4_6      144
30     q7_4      q7_4_7      96
31     q7_4      q7_4_8      352
32     q7_5      q7_5_1      645
33     q7_5      q7_5_2      173
34     q7_5      q7_5_3      381
35     q7_5      q7_5_4      791
36     q7_5      q7_5_5      437
37     q7_5      q7_5_6      239
38     q7_5      q7_5_7      124
39     q7_5      q7_5_8      304
```

```
labels_df = pd.read_csv('labels.csv')
labels_df
```

	Question ID	Question Code	Question	Base option	Precode	Subcode	Option suffix
0	s2_359	s2	What country do you live in currently?	Bulgaria	359	NaN	NaN
1	s2_420	s2	What country do you live in currently?	Czech Republic	420	NaN	NaN
2	s2_36	s2	What country do you live in currently?	Hungary	36	NaN	NaN
3	s2_48	s2	What country do you live in currently?	Poland	48	NaN	NaN
4	s2_40	s2	What country do you live in currently?	Romania	40	NaN	NaN
...	...	...	...	...	...	...	...
98	q7_5_4	q7	For each of the purposes listed below, which o...	To help me find the best brand/product relevan...	5	4.0	Price comparison apps or websites
99	q7_5_5	q7	For each of the purposes listed below, which o...	To help me find the best brand/product relevan...	5	5.0	Online video websites(e.g. Youtube)
100	q7_5_6	q7	For each of the purposes listed below, which o...	To help me find the best brand/product relevan...	5	6.0	Influencer blogs / vlogs
101	q7_5_7	q7	For each of the purposes listed below, which o...	To help me find the best brand/product relevan...	5	7.0	Other

```
labels_df.columns
```

```
Index(['Question ID', 'Question Code', 'Question', 'Base option', 'Precode', 'Subcode', 'Option suffix'], dtype='object')
```

```
labels_df['Question ID'].unique()
```

```
array(['s2_359', 's2_420', 's2_36', 's2_48', 's2_40', 'q2_1', 'q2_2',
      'q3_18', 'q3_19', 'q3_20', 'q3_21', 'q3_22', 'q3_23', 'q3_24',
      'q3_25', 'q3_26', 'q3_27', 'q3_28', 'q3_29', 'q3_30', 'q3_31',
      'q3_32', 'q3_33', 'q3_34', 'q3_35', 'q3_36', 'q3_37', 'q3_38',
      'q3_39', 'q3_40', 'q3_41', 'q3_42', 'q3_43', 'q3_44', 'q3_45',
      'q3_46', 'q3_47', 'q3_48', 'q3_49', 'q3_50', 'q3_51', 'q3_52',
      'q3_53', 'q3_54', 'q3_55', 'q3_56', 'q3_57', 'q3_58', 'q3_59',
      'q3_60', 'q3_61', 'q3_62', 'q3_63', 'q3_64', 'q3a_1', 'q3a_2',
      'q3a_3', 'q3a_4', 'q3a_5', 'q3a_6', 'q48_1', 'q48_2', 'q48_3',
      'q7_1_1', 'q7_1_2', 'q7_1_3', 'q7_1_4', 'q7_1_5', 'q7_1_6',
      'q7_1_7', 'q7_1_8', 'q7_2_1', 'q7_2_2', 'q7_2_3', 'q7_2_4',
      'q7_2_5', 'q7_2_6', 'q7_2_7', 'q7_2_8', 'q7_3_1', 'q7_3_2',
      'q7_3_3', 'q7_3_4', 'q7_3_5', 'q7_3_6', 'q7_3_7', 'q7_3_8',
      'q7_4_1', 'q7_4_2', 'q7_4_3', 'q7_4_4', 'q7_4_5', 'q7_4_6',
      'q7_4_7', 'q7_4_8', 'q7_5_1', 'q7_5_2', 'q7_5_3', 'q7_5_4',
      'q7_5_5', 'q7_5_6', 'q7_5_7', 'q7_5_8'], dtype=object)
```

```
labels_df['Question Code'].unique()
```

```
array(['s2', 'q2', 'q3', 'q3a', 'q48', 'q7'], dtype=object)
```

```
labels_df['Question'].unique()
```

```
array(['What country do you live in currently?',
      'Which of the following best describes your gender?',
      'How old are you?', 'Age groups',
      'Which best describes the area where you live?',
      'For each of the purposes listed below, which of these ONLINE touchpoints have ever influenced your buying decision? Please select all that apply per row'],
      dtype=object)
```

```
col_groups = {
    code: group_df
    for code, group_df in labels_df.groupby('Question Code')
}
col_groups['s2']
```

	Question ID	Question Code	Question	Base option	Precode	Subcode	Option suffix
0	s2_359	s2	What country do you live in currently?	Bulgaria	359	NaN	NaN
1	s2_420	s2	What country do you live in currently?	Czech Republic	420	NaN	NaN
2	s2_36	s2	What country do you live in currently?	Hungary	36	NaN	NaN
3	s2_48	s2	What country do you live in currently?	Poland	48	NaN	NaN
4	s2_40	s2	What country do you live in currently?	Romania	40	NaN	NaN

```
col_groups['q2']
```

	Question ID	Question Code	Question	Base option	Precode	Subcode	Option suffix
5	q2_1	q2	Which of the following best describes your gen...	Male	1	NaN	NaN
6	q2_2	q2	Which of the following best describes your gen...	Female	2	NaN	NaN

```
col_groups['q7']
```


	Question ID	Question Code	Question	Base option	Precode	Subcode	Option suffix
63	q7_1_1	q7	For each of the purposes listed below, which o...	Help me discover a new brand, product or service	1	1.0	Search engines
64	q7_1_2	q7	For each of the purposes listed below, which o...	Help me discover a new brand, product or service	1	2.0	Social media platforms(e.g. Facebook, Instagram)
65	q7_1_3	q7	For each of the purposes listed below, which o...	Help me discover a new brand, product or service	1	3.0	Product / brand apps or websites
66	q7_1_4	q7	For each of the purposes listed below, which o...	Help me discover a new brand, product or service	1	4.0	Price comparison apps or websites
67	q7_1_5	q7	For each of the purposes listed below, which o...	Help me discover a new brand, product or service	1	5.0	Online video websites(e.g. Youtube)
68	q7_1_6	q7	For each of the purposes listed below, which o...	Help me discover a new brand, product or service	1	6.0	Influencer blogs / vlogs
69	q7_1_7	q7	For each of the purposes listed below, which o...	Help me discover a new brand, product or service	1	7.0	Other
70	q7_1_8	q7	For each of the purposes listed below, which o...	Help me discover a new brand, product or service	1	8.0	NaN
71	q7_2_1	q7	For each of the purposes listed below, which o...	Help me determine the quality	2	1.0	Search engines
72	q7_2_2	q7	For each of the purposes listed below, which o...	Help me determine the quality	2	2.0	Social media platforms(e.g. Facebook, Instagram)

```
col_groups['q7'].groupby('Base option')['Subcode'].agg(  
    total_responses='count',  
    null_responses=lambda x: x.isnull().sum()  
)
```

75	q7_2_5	q7	For each of the purposes listed below, which o...	total_responses	8	0	2	5.0	Online video websites(e.g. Youtube)
76	q7_2_6	q7	For each of the purposes listed below, which o...	total_responses	8	0	2	6.0	Influencer blogs / vlogs
77	q7_2_7	q7	For each of the purposes listed below, which o...	total_responses	8	0	2	7.0	Other
78	q7_2_8	q7	For each of the purposes listed below, which o...	total_responses	8	0	2	8.0	NaN
79	q7_2_9	q7	For each of the purposes listed below, which o...	total_responses	8	0	3	1.0	Search engines

```
for code in labels_df['Question Code'].unique():  
    print(col_groups[code])
```

81	q7_3_3	q7	For each of the purposes listed below, which o...	To give my confidence in my choice	3	3.0	Product / brand apps or websites
0	s2_359	s2	What country do you live in currently?	To give my confidence in my choice	3	4.0	Price comparison apps or websites
1	s2_420	s2	What country do you live in currently?	To give my confidence in my choice	3	5.0	Online video websites(e.g. Youtube)
2	s2_36	s2	What country do you live in currently?	To give my confidence in my choice	3	6.0	Influencer blogs / vlogs
3	s2_48	s2	What country do you live in currently?	To give my confidence in my choice	3	7.0	Other
4	s2_40	s2	What country do you live in currently?	To give my confidence in my choice	3	8.0	NaN
0	s1_39	s1	For each of the purposes listed below, which o...	To give my confidence in my choice	3	6.0	Influencer blogs / vlogs
1	Czech Republic	420	For each of the purposes listed below, which o...	To give my confidence in my choice	3	7.0	Other
2	Hungary	36	For each of the purposes listed below, which o...	To give my confidence in my choice	3	8.0	NaN
3	Poland	489	For each of the purposes listed below, which o...	To give my confidence in my choice	3	9.0	NaN
4	Romania	40	For each of the purposes listed below, which o...	To give my confidence in my choice	3	10.0	NaN
5	q7_3_8	q7	For each of the purposes listed below, which o...	To give my confidence in my choice	3	8.0	NaN
6	q2_2	q2	For each of the purposes listed below, which o...	To give my confidence in my choice	3	8.0	NaN
87	q7_4_1	q7	For each of the purposes listed below, which o...	To reassure me it's the right price	4	1.0	Search engines
5	Which of the following best describes your gen...		For each of the purposes listed below, which o...	To reassure me it's the right price	4	2.0	Social media platforms(e.g. Facebook, Instagram)
6	Which of the following best describes your gen...		For each of the purposes listed below, which o...	To reassure me it's the right price	4	3.0	Product / brand apps or websites
5	89	NaN	For each of the purposes listed below, which o...	To reassure me it's the right price	4	3.0	Product / brand apps or websites
6	NaN	NaN	For each of the purposes listed below, which o...	To reassure me it's the right price	4	3.0	Product / brand apps or websites
7	90	q3_18	For each of the purposes listed below, which o...	To reassure me it's the right price	4	4.0	Price comparison apps or websites
8	q3_19	q3	How old are you?	To reassure me it's the right price	4	4.0	Price comparison apps or websites
9	q3_20	q3	How old are you?	To reassure me it's the right price	4	5.0	Online video websites(e.g. Youtube)
10	91	q3_21	For each of the purposes listed below, which o...	To reassure me it's the right price	4	5.0	Online video websites(e.g. Youtube)

11	q3_22	q3	How old are you?	22	To reassure me it's the right	NaN			
12	q3_23	q3	How old are you?	23	To reassure me it's the right	NaN	4	6.0	Influencer blogs / vlogs
13	q3_24	q3	How old are you?	24	NaN	NaN			
14	q3_25	q3	For each of the purposes listed below, which o...	25	To reassure me it's the right	NaN			
15	q3_26	q3	How old are you?	26	To reassure me it's the right	NaN	4	7.0	Other
16	q3_27	q3	How old are you?	27	NaN	NaN			
17	q3_28	q3	For each of the purposes listed below, which o...	28	To reassure me it's the right	NaN			
18	q3_29	q3	How old are you?	29	To reassure me it's the right	NaN	4	8.0	NaN
19	q3_30	q3	How old are you?	30	NaN	NaN			
20	q3_31	q3	For each of the purposes listed below, which o...	31	To help me find the best brand/product relevan...	NaN	5	1.0	Search engines
21	q3_32	q3	How old are you?	32	NaN	NaN			
22	q3_33	q3	How old are you?	33	NaN	NaN			
23	q3_34	q3	For each of the purposes listed below, which o...	34	To help me find the best brand/product relevan...	NaN	5	2.0	Social media platforms(e.g. Facebook, Instagram)
24	q3_35	q3	How old are you?	35	NaN	NaN			
25	q3_36	q3	How old are you?	36	NaN	NaN			
26	q3_37	q3	For each of the purposes listed below, which o...	37	To help me find the best brand/product relevan...	NaN	5	3.0	Product / brand apps or websites
27	q3_38	q3	How old are you?	38	NaN	NaN			
28	q3_39	q3	For each of the purposes listed below, which o...	39	To help me find the best brand/product relevan...	NaN			
29	q3_40	q3	How old are you?	40	To help me find the best brand/product relevan...	NaN	5	4.0	Price comparison apps or websites
30	q3_41	q3	How old are you?	41	NaN	NaN			
31	q3_42	q3	For each of the purposes listed below, which o...	42	To help me find the best brand/product relevan...	NaN	5	5.0	Online video websites(e.g. Youtube)
32	q3_43	q3	How old are you?	43	NaN	NaN			
33	q3_44	q3	How old are you?	44	NaN	NaN			
34	q3_45	q3	For each of the purposes listed below, which o...	45	To help me find the best brand/product relevan...	NaN	5	6.0	Influencer blogs / vlogs
35	q3_46	q3	How old are you?	46	NaN	NaN			
36	q3_47	q3	How old are you?	47	NaN	NaN			
37	q3_48	q3	For each of the purposes listed below, which o...	48	To help me find the best brand/product relevan...	NaN	5	7.0	Other
38	q3_49	q3	How old are you?	49	NaN	NaN			
39	q3_50	q3	How old are you?	50	NaN	NaN			
40	q3_51	q3	For each of the purposes listed below, which o...	51	To help me find the best brand/product relevan...	NaN			

labels_df.isnull().sum().sort_values(ascending=False)

	0
Option suffix	68
Subcode	63
Question ID	0
Question	0
Question Code	0
Precode	0
Base option	0

dtype: int64

labels_df[labels_df.isnull().any(axis=1)]

	Question ID	Question Code	Question	Base option	Precode	Subcode	Option suffix
0	s2_359	s2	What country do you live in currently?	Bulgaria	359	NaN	NaN
1	s2_420	s2	What country do you live in currently?	Czech Republic	420	NaN	NaN
2	s2_36	s2	What country do you live in currently?	Hungary	36	NaN	NaN
3	s2_48	s2	What country do you live in currently?	Poland	48	NaN	NaN
4	s2_40	s2	What country do you live in currently?	Romania	40	NaN	NaN
...	...	...	...	...	...	...	...
70	q7_1_8	q7	For each of the purposes listed below, which o...	Help me discover a new brand, product or service	1	8.0	NaN
78	q7_2_8	q7	For each of the purposes listed below, which o...	Help me determine the quality	2	8.0	NaN
86	q7_3_8	q7	For each of the purposes listed below, which o...	To give my confidence in my choice	3	8.0	NaN
94	q7_4_8	q7	For each of the purposes listed below, which o...	To reassure me it's the right price	4	8.0	NaN
102	q7_5_8	q7	For each of the purposes listed below, which o...	To help me find the best brand/product relevan...	5	8.0	NaN

68 rows × 7 columns

```
null_counts = labels_df[labels_df['Option suffix'].isnull()].groupby('Question Code').size()
print(null_counts)
```

```
Question Code
q2          2
q3         47
q3a         6
q48         3
q7          5
s2          5
dtype: int64
```

```
null_counts = labels_df[labels_df['Subcode'].isnull()].groupby('Question Code').size()
print(null_counts)
```

```
Question Code
q2          2
q3         47
q3a         6
q48         3
s2          5
dtype: int64
```

```
null_counts = labels_df[labels_df['Subcode'].isnull()].groupby('Question').size()
print(null_counts)
```

```
Question
Age groups                                6
How old are you?                          47
What country do you live in currently?     5
Which best describes the area where you live? 3
Which of the following best describes your gender? 2
dtype: int64
```

```
null_counts = labels_df[labels_df['Subcode'].isnull()].groupby('Question ID').size()
print(null_counts)
```

```
Question ID
q2_1      1
q2_2      1
q3_18     1
q3_19     1
q3_20     1
..
s2_359    1
s2_36     1
s2_40     1
s2_420    1
s2_48     1
Length: 63, dtype: int64
```

```
#labels_df[labels_df.isnull().any(axis=1)]
# Filter rows with null answer_code
null_entries = labels_df[labels_df['Subcode'].isnull()]
# Group by question_code and display
for question, group in null_entries.groupby('Question'):
    print(f"\n=== Null Responses for Question: {question} ===")
    display(group)
```


=== Null Responses for Question: Age groups ===

	Question ID	Question Code	Question	Base option	Precode	Subcode	Option suffix
54	q3a_1	q3a	Age groups	18-19	1	NaN	NaN
55	q3a_2	q3a	Age groups	20-29	2	NaN	NaN
56	q3a_3	q3a	Age groups	30-39	3	NaN	NaN
57	q3a_4	q3a	Age groups	40-49	4	NaN	NaN
58	q3a_5	q3a	Age groups	50-59	5	NaN	NaN
59	q3a_6	q3a	Age groups	60-64	6	NaN	NaN

=== Null Responses for Question: How old are you? ===

	Question ID	Question Code	Question	Base option	Precode	Subcode	Option suffix
7	q3_18	q3	How old are you?		18	18	NaN
8	q3_19	q3	How old are you?		19	19	NaN
9	q3_20	q3	How old are you?		20	20	NaN
10	q3_21	q3	How old are you?		21	21	NaN
11	q3_22	q3	How old are you?		22	22	NaN
12	q3_23	q3	How old are you?		23	23	NaN
13	q3_24	q3	How old are you?		24	24	NaN
14	q3_25	q3	How old are you?		25	25	NaN
15	q3_26	q3	How old are you?		26	26	NaN
16	q3_27	q3	How old are you?		27	27	NaN
17	q3_28	q3	How old are you?		28	28	NaN
18	q3_29	q3	How old are you?		29	29	NaN
19	q3_30	q3	How old are you?		30	30	NaN
20	q3_31	q3	How old are you?		31	31	NaN
21	q3_32	q3	How old are you?		32	32	NaN
22	q3_33	q3	How old are you?		33	33	NaN
23	q3_34	q3	How old are you?		34	34	NaN

```
raw_df = pd.read_csv('live_test.csv')
raw_df
```

26	q3_37	q3	How old are you?		37	37	NaN
27	q3_38	q3	How old are you?		38	38	NaN
28	q3_39	q3	How old are you?		39	39	NaN
29	q3_40	q3	How old are you?		40	40	NaN
30	q3_41	q3	How old are you?		41	41	NaN
31	q3_42	q3	How old are you?		42	42	NaN
32	q3_43	q3	How old are you?		43	43	NaN
33	q3_44	q3	How old are you?		44	44	NaN
34	q3_45	q3	How old are you?		45	45	NaN
35	q3_46	q3	How old are you?		46	46	NaN
36	q3_47	q3	How old are you?		47	47	NaN
37	q3_48	q3	How old are you?		48	48	NaN
38	q3_49	q3	How old are you?		49	49	NaN
39	q3_50	q3	How old are you?		50	50	NaN
40	q3_51	q3	How old are you?		51	51	NaN
41	q3_52	q3	How old are you?		52	52	NaN
42	q3_53	q3	How old are you?		53	53	NaN

43	q3_34	q3	How old are you?				34	34	NaN	NaN						
44	respondent_id	q2_1	q2_2	q2_3	q3_18	q3_19	q3_20	q3_21	q3_22	q3_23	...	s2_420	s2_48	Q19_1_TEXT	Q22_1_TEXT	Q24_1
	q3_55	q3	How old are you?				55	55	NaN	NaN		NaN				
0	jtn-31kgg8psdBCqYD6	1	0	0	0	0	0	0	0	0	...	1	0	NaN	NaN	
46	q3_57	q3	How old are you?				57	57	NaN	NaN		NaN				
1	5FQ679JNrH6g7dT	1	0	0	0	0	0	0	0	0	...	1	0	NaN	NaN	
47	q3_58	q3	How old are you?				58	58	NaN	NaN		NaN				
2	jtn-22853lleE3oxpXj	1	0	0	0	0	0	0	0	0	...	0	0	NaN	NaN	
48	q3_60	q3	How old are you?				60	60	NaN	NaN		NaN	1	0	NaN	NaN
...	1491R1Uo7f7tb0f	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
4	jtn-2UaYA52mtFm5pF3	1	0	0	0	0	0	0	0	0	...	0	0	NaN	NaN	
52	q3_63	...	...	...	...	...	63	63	NaN	NaN	...	NaN	...	...	...	...
3089	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	0	1	0	0	0	0	0	0	0	...	0	1	NaN	NaN	
3090	Question ID	Question Code	1	0	0	Question	Base option	Precode	Subcode	Option suffix	NaN	NaN				
...	vG9F16NqfRwxVLx	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
3091	jtn-4guyjfQ0PtcYI91	0	1	0	0	0	0	0	0	0	...	1	0	NaN	NaN	
1	sZ_4ZU	sZ	vwnat country do you live in currently?				Czech Republic	4ZU	NaN	NaN		NaN				
3092	jtn-za370CyyvD9fcoZn	1	0	0	0	0	0	0	0	0	...	NaN	0	NaN	NaN	NaN
2	...	s2	What country do you live in currently?				Hungary	36	NaN	NaN		NaN				
3093	jtn-uqgmsyrCPGOMYf8	1	0	0	0	0	0	0	0	0	...	1	0	NaN	NaN	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
3094 rows × 67 columns																
=== Null Responses for Question: Which best describes the area where you live? ===																

```
Question ID Question Code Question Base option Precode Subcode Option suffix
q7_cols = [col for col in raw_df.columns if col.startswith('q7_')]
long_q7 = raw_df.melt(id_vars='respondent_id', value_vars=q7_cols,
                      var_name='question_id', value_name='response')
long_q7
```

=== Null Responses for Question: Which of the following best describes your gender? ===																
0	jtn-31kgg8psdBCqYD6	q7_1	8	on	Base option	Precode	Subcode	Option suffix								
5 1	q2_1	jtn-5FQ679JNrH6g7dT	q7_1	4	...	Male	1	NaN	NaN							
2		jtn-22853lleE3oxpXj	q7_1	4	...	Female	2	NaN	NaN							
3		jtn-1491R1Uo7f7tb0f	q7_1	8												
4		jtn-2UaYA52mtFm5pF3	q7_1	5												
...		...	...	...												
15465		toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	q7_5	4												
15466		jtn-vG9F16NqfRwxVLx	q7_5	8												
15467		jtn-4guyjfQ0PtcYI91	q7_5	1												
15468		jtn-za370CyyvD9fcoZn	q7_5	6												
15469		jtn-uqgmsyrCPGOMYf8	q7_5	1												
15470 rows × 3 columns																

```
#labels_df[labels_df.isnull().any(axis=1)]
# Filter rows with null answer_code
null_entries = long_q7[long_q7.isnull()]
# Group by question_code and display
for question, group in null_entries.groupby('question_id'):
    print(f"\n=== Null Responses for Question: {question} ===")
    display(group)
```

```
long_q7.groupby('question_id').agg('count')
```

	respondent_id	response
question_id		
q7_1	3094	3094
q7_2	3094	3094
q7_3	3094	3094
q7_4	3094	3094
q7_5	3094	3094

df_q7_long

	respondent_id	question_base	precode	question_id	description
0	jtn-31kgg8psdBCqYD6	q7_1	8	q7_1_8	Help me discover a new brand, product or service
1	jtn-5FQ679JNrH6g7dT	q7_1	4	q7_1_4	Help me discover a new brand, product or service
2	jtn-22853lleE3oxpXj	q7_1	4	q7_1_4	Help me discover a new brand, product or service
3	jtn-1491R1Uo7f7tb0f	q7_1	8	q7_1_8	Help me discover a new brand, product or service
4	jtn-2UaYA52mtFm5pF3	q7_1	5	q7_1_5	Help me discover a new brand, product or service
...	...	...	...	...	...
15465	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	q7_5	4	q7_5_4	To help me find the best brand/product relevan...
15466	jtn-vG9F16NqfRwxVLx	q7_5	8	q7_5_8	To help me find the best brand/product relevan...
15467	jtn-4guyjfQ0PtcYI91	q7_5	1	q7_5_1	To help me find the best brand/product relevan...
15468	jtn-za370CyvD9fcoZn	q7_5	6	q7_5_6	To help me find the best brand/product relevan...
15469	jtn-uqgmsyrCPGOMYf8	q7_5	1	q7_5_1	To help me find the best brand/product relevan...

15470 rows × 5 columns

df_q7_long.groupby('question_id')['question_base'].agg('count')

question_base

question_id

q7_1_1	639
q7_1_2	283
q7_1_3	372
q7_1_4	539
q7_1_5	487
q7_1_6	315
q7_1_7	121
q7_1_8	338
q7_2_1	514
q7_2_2	211
q7_2_3	387
q7_2_4	641
q7_2_5	494
q7_2_6	270
q7_2_7	157
q7_2_8	420
q7_3_1	490
q7_3_2	181
q7_3_3	323
q7_3_4	743
q7_3_5	446
q7_3_6	265
q7_3_7	160
q7_3_8	486
q7_4_1	517
q7_4_2	158
q7_4_3	248
q7_4_4	1334
q7_4_5	245
q7_4_6	144
q7_4_7	96
q7_4_8	352
q7_5_1	645
q7_5_2	173
q7_5_3	381
q7_5_4	791
q7_5_5	437
q7_5_6	239
q7_5_7	124
q7_5_8	304

dtype: int64

```
df_q7_long.groupby('question_id').agg('count')
```

	respondent_id	question_base	precode	description
question_id				
q7_1_1	639	639	639	639
q7_1_2	283	283	283	283
q7_1_3	372	372	372	372
q7_1_4	539	539	539	539
q7_1_5	487	487	487	487
q7_1_6	315	315	315	315
q7_1_7	121	121	121	121
q7_1_8	338	338	338	338
q7_2_1	514	514	514	514
q7_2_2	211	211	211	211
q7_2_3	387	387	387	387
q7_2_4	641	641	641	641
q7_2_5	494	494	494	494
q7_2_6	270	270	270	270
q7_2_7	157	157	157	157
q7_2_8	420	420	420	420
q7_3_1	490	490	490	490
q7_3_2	181	181	181	181
q7_3_3	323	323	323	323
q7_3_4	743	743	743	743
q7_3_5	446	446	446	446
q7_3_6	265	265	265	265
q7_3_7	160	160	160	160
q7_3_8	486	486	486	486
q7_4_1	517	517	517	517
q7_4_2	158	158	158	158
q7_4_3	248	248	248	248
q7_4_4	1334	1334	1334	1334
q7_4_5	245	245	245	245
q7_4_6	144	144	144	144
q7_4_7	96	96	96	96
q7_4_8	352	352	352	352
q7_5_1	645	645	645	645
q7_5_2	173	173	173	173
q7_5_3	381	381	381	381
q7_5_4	791	791	791	791
q7_5_5	437	437	437	437
q7_5_6	239	239	239	239
q7_5_7	124	124	124	124
q7_5_8	304	304	304	304

raw_df.columns

Index(['respondent_id', 'q2_1', 'q2_2', 'q2_3', 'q3_18', 'q3_19', 'q3_20',
 'q3_21', 'q3_22', 'q3_23', 'q3_24', 'q3_25', 'q3_26', 'q3_27', 'q3_28',
 'q3_29', 'q3_30', 'q3_31', 'q3_32', 'q3_33', 'q3_34', 'q3_35', 'q3_36',
 'q3_37', 'q3_38', 'q3_39', 'q3_40', 'q3_41', 'q3_42', 'q3_43', 'q3_44',

```
'q3_45', 'q3_46', 'q3_47', 'q3_48', 'q3_49', 'q3_50', 'q3_51', 'q3_52',
'q3_53', 'q3_54', 'q3_55', 'q3_56', 'q3_57', 'q3_58', 'q3_59', 'q3_60',
'q3_61', 'q3_62', 'q3_63', 'q3_64', 'q48_1', 'q48_2', 'q48_3', 's2_359',
's2_36', 's2_40', 's2_420', 's2_48', 'Q19_1_TEXT', 'Q22_1_TEXT',
'Q24_6_TEXT', 'q7_1', 'q7_2', 'q7_3', 'q7_4', 'q7_5'],
dtype='object')
```

```
raw_df.info()
```

```
13 q3_27      3094 non-null  int64
14 q3_28      3094 non-null  int64
15 q3_29      3094 non-null  int64
16 q3_30      3094 non-null  int64
17 q3_31      3094 non-null  int64
18 q3_32      3094 non-null  int64
19 q3_33      3094 non-null  int64
20 q3_34      3094 non-null  int64
21 q3_35      3094 non-null  int64
22 q3_36      3094 non-null  int64
23 q3_37      3094 non-null  int64
24 q3_38      3094 non-null  int64
25 q3_39      3094 non-null  int64
26 q3_40      3094 non-null  int64
27 q3_41      3094 non-null  int64
28 q3_42      3094 non-null  int64
29 q3_43      3094 non-null  int64
30 q3_44      3094 non-null  int64
31 q3_45      3094 non-null  int64
32 q3_46      3094 non-null  int64
33 q3_47      3094 non-null  int64
34 q3_48      3094 non-null  int64
35 q3_49      3094 non-null  int64
36 q3_50      3094 non-null  int64
37 q3_51      3094 non-null  int64
38 q3_52      3094 non-null  int64
39 q3_53      3094 non-null  int64
40 q3_54      3094 non-null  int64
41 q3_55      3094 non-null  int64
42 q3_56      3094 non-null  int64
43 q3_57      3094 non-null  int64
44 q3_58      3094 non-null  int64
45 q3_59      3094 non-null  int64
46 q3_60      3094 non-null  int64
47 q3_61      3094 non-null  int64
48 q3_62      3094 non-null  int64
49 q3_63      3094 non-null  int64
50 q3_64      3094 non-null  int64
51 q48_1      3094 non-null  int64
52 q48_2      3094 non-null  int64
53 q48_3      3094 non-null  int64
54 s2_359     3094 non-null  int64
55 s2_36      3094 non-null  int64
56 s2_40      3094 non-null  int64
57 s2_420     3094 non-null  int64
58 s2_48      3094 non-null  int64
59 Q19_1_TEXT  31 non-null    object
60 Q22_1_TEXT  17 non-null    object
61 Q24_6_TEXT  4 non-null     object
62 q7_1       3094 non-null  int64
63 q7_2       3094 non-null  int64
64 q7_3       3094 non-null  int64
65 q7_4       3094 non-null  int64
66 q7_5       3094 non-null  int64
67 country    3094 non-null  object
68 area       3094 non-null  object
dtypes: int64(63), object(6)
memory usage: 1.6+ MB
```

```
def get_country(df):
    country_cols = [col for col in df.columns if col.startswith("s2_")]
    return df[country_cols].idxmax(axis=1).map(lambda x: x.split('_')[1])

get_country(raw_df)
```

	0
0	420
1	420
2	36
3	420
4	36
...	...
3089	48
3090	359
3091	420
3092	36
3093	420

3094 rows × 1 columns

dtype: object

```
raw_df[(raw_df['q2_1'] == 1)].shape
```

(1549, 69)

```
raw_df['gender'].value_counts()[[0]] #.plot(kind='bar')
```

/tmp/ipython-input-158-1156451153.py:1: FutureWarning: Series.__getitem__ treating keys as positions is deprecated. In a future version this will raise a KeyError. The following keys are being treated as positions: [0]
raw_df['gender'].value_counts()[[0]] #.plot(kind='bar')

count	
gender	
Male	1549

dtype: int64

df_q7_long
df_q7_long_mapped

	respondent_id	question_base	precode	question_id	description
0	jtn-31kgg8psdBCqYD6	q7_1	8	q7_1_8	Help me discover a new brand, product or service
1	jtn-5FQ679JNrH6g7dT	q7_1	4	q7_1_4	Help me discover a new brand, product or service
2	jtn-22853lleE3oxpXj	q7_1	4	q7_1_4	Help me discover a new brand, product or service
3	jtn-1491R1Uo7f7tb0f	q7_1	8	q7_1_8	Help me discover a new brand, product or service
4	jtn-2UaYA52mtFm5pF3	q7_1	5	q7_1_5	Help me discover a new brand, product or service
...	...	...	...	...	...
15465	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	q7_5	4	q7_5_4	To help me find the best brand/product relevan...
15466	jtn-vG9F16NqfRwxVLx	q7_5	8	q7_5_8	To help me find the best brand/product relevan...
15467	jtn-4guyjfQ0PtYI91	q7_5	1	q7_5_1	To help me find the best brand/product relevan...
15468	jtn-za370CyvD9fcoZn	q7_5	6	q7_5_6	To help me find the best brand/product relevan...
15469	jtn-uqgmsyrCPGOMYf8	q7_5	1	q7_5_1	To help me find the best brand/product relevan...

15470 rows × 5 columns

```
#df.groupby(['gender', 'age_group']).size().reset_index(name='count')
```

```
raw_df.groupby('gender')['total_selected_q7'].mean()
```

total_selected_q7	
gender	
Female	4.319741

df[df.duplicated(subset=['respondent_id'], keep=False)].groupby('respondent_id').size()

dtype: float64	
0	
respondent_id	
acuity-AKP-72Cxz1721419664aShUu	2
acuity-AKP-7480ab55-aed7-457c-95ca-20fbd562d6df	2
jtn-4guyjfQ0PtcYI91	2
jtn-X3x1fr7Ea1cwa2o	2
jtn-hkhB0CD7llkaL75	2
jtn-nffYkv190v2Z34A	2
jtn-uqgmsyrCPGOMYf8	2
jtn-vG9F16NqfRwxVLx	2
jtn-za370CyvD9fcoZn	2
talkonline-66b2a870-abe4-11ee-8f68-b7b583e7c05a	2
toluna-5729c24e-851e-4f6f-91e6-e7df8c6dc5f	2

dtype: int64

df[df.duplicated()]['respondent_id']

respondent_id	
570	jtn-nffYkv190v2Z34A
3084	jtn-hkhB0CD7llkaL75
3085	acuity-AKP-72Cxz1721419664aShUu
3086	talkonline-66b2a870-abe4-11ee-8f68-b7b583e7c05a
3087	jtn-X3x1fr7Ea1cwa2o
3088	acuity-AKP-7480ab55-aed7-457c-95ca-20fbd562d6df
3089	toluna-5729c24e-851e-4f6f-91e6-e7df8c6dc5f
3090	jtn-vG9F16NqfRwxVLx
3091	jtn-4guyjfQ0PtcYI91
3092	jtn-za370CyvD9fcoZn
3093	jtn-uqgmsyrCPGOMYf8

dtype: object

df[df.duplicated(keep='last')]['respondent_id']

	respondent_id
354	jtn-vG9F16NqfRwxVLx
408	jtn-uqgmsyrCPGOMYf8
511	jtn-X3x1fr7Ea1cwa2o
569	jtn-nffYkv190v2Z34A
790	toluna-5729c24e-851e-4f6f-91e6-e7df8c6dc5f
1505	jtn-hkhB0CD7llkaL75
1565	acuity-AKP-7480ab55-aed7-457c-95ca-20fd562d6df
1651	jtn-4guyjfQ0PtcYl91
1814	talkonline-66b2a870-abe4-11ee-8f68-b7b583e7c05a
2398	acuity-AKP-72Cxz1721419664aShUu
2858	jtn-za370CyvD9fcoZn

dtype: object

```
df.loc[df.duplicated(keep=False)]
```

	respondent_id	q2_1	q2_2	q2_3	q3_18	q3_19	q3_20	q3_21	q3_22	q3_23	...	Q24_6_TEXT	q7_1	q7_2	q7_3	q7_4	q7_5
354	jtn-vG9F16NqfRwxVLx	0	1	0	0	0	0	0	0	0	...	NaN	6	3	6	8	
408	jtn-uqgmsyrCPGOMYf8	1	0	0	0	0	0	0	0	0	...	NaN	4	4	4	4	
511	jtn-X3x1fr7Ea1cwa2o	0	1	0	0	0	0	0	0	0	...	NaN	3	3	4	4	
569	jtn-nffYkv190v2Z34A	1	0	0	0	0	0	0	0	0	...	NaN	3	3	3	3	
570	jtn-nffYkv190v2Z34A	1	0	0	0	0	0	0	0	0	...	NaN	3	3	3	3	
790	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	0	1	0	0	0	0	0	0	0	...	NaN	4	4	4	4	
1505	jtn-hkhB0CD7IlkaL75	1	0	0	0	0	0	0	0	0	...	NaN	4	2	4	4	
1565	acuity-AKP-7480ab55-aed7-457c-95ca-20fbd562d6df	0	1	0	0	0	0	0	0	0	...	NaN	3	3	3	3	
1651	jtn-4guyjfQ0PtcYI91	0	1	0	0	0	0	0	0	0	...	NaN	1	1	1	8	
1814	talkonline-66b2a870-abe4-11ee-8f68-b7b583e7c05a	0	1	0	0	0	0	0	0	0	...	NaN	1	7	1	1	
2398	acuity-AKP-72Cxz1721419664aShUu	0	1	0	0	0	0	0	0	0	...	NaN	1	2	2	4	
2858	jtn-za370CyvD9fcoZn	1	0	0	0	0	0	0	0	0	...	NaN	6	5	6	5	
3084	jtn-hkhB0CD7IlkaL75	1	0	0	0	0	0	0	0	0	...	NaN	4	2	4	4	
3085	acuity-AKP-72Cxz1721419664aShUu	0	1	0	0	0	0	0	0	0	...	NaN	1	2	2	4	
3086	talkonline-66b2a870-abe4-11ee-8f68-b7b583e7c05a	0	1	0	0	0	0	0	0	0	...	NaN	1	7	1	1	
3087	jtn-X3x1fr7Ea1cwa2o	0	1	0	0	0	0	0	0	0	...	NaN	3	3	4	4	
3088	acuity-AKP-7480ab55-aed7-457c-95ca-20fbd562d6df	0	1	0	0	0	0	0	0	0	...	NaN	3	3	3	3	
3089	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	0	1	0	0	0	0	0	0	0	...	NaN	4	4	4	4	
3090	jtn-vG9F16NqfRwxVLx	0	1	0	0	0	0	0	0	0	...	NaN	6	3	6	8	
3091	jtn-4guyjfQ0PtcYI91	0	1	0	0	0	0	0	0	0	...	NaN	1	1	1	8	
3092	jtn-za370CyvD9fcoZn	1	0	0	0	0	0	0	0	0	...	NaN	6	5	6	5	
3093	jtn-uqgmsyrCPGOMYf8	1	0	0	0	0	0	0	0	0	...	NaN	4	4	4	4	

22 rows × 17 columns

```
df[df.duplicated(keep=False)]
```

	respondent_id	q2_1	q2_2	q2_3	q3_18	q3_19	q3_20	q3_21	q3_22	q3_23	...	Q24_6_TEXT	q7_1	q7_2	q7_3	q7_4	q7_5
354	jtn-vG9F16NqfRwxVLx	0	1	0	0	0	0	0	0	0	...	NaN	6	3	6	8	
408	jtn-uqgmsyrCPGOMYf8	1	0	0	0	0	0	0	0	0	...	NaN	4	4	4	4	
511	jtn-X3x1fr7Ea1cwa2o	0	1	0	0	0	0	0	0	0	...	NaN	3	3	4	4	
569	jtn-nffYkv190v2Z34A	1	0	0	0	0	0	0	0	0	...	NaN	3	3	3	3	
570	jtn-nffYkv190v2Z34A	1	0	0	0	0	0	0	0	0	...	NaN	3	3	3	3	
790	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	0	1	0	0	0	0	0	0	0	...	NaN	4	4	4	4	
1505	jtn-hkhB0CD7IlkaL75	1	0	0	0	0	0	0	0	0	...	NaN	4	2	4	4	
1565	acuity-AKP-7480ab55-aed7-457c-95ca-20fbd562d6df	0	1	0	0	0	0	0	0	0	...	NaN	3	3	3	3	
1651	jtn-4guyjfQ0PtYI91	0	1	0	0	0	0	0	0	0	...	NaN	1	1	1	8	
1814	talkonline-66b2a870-abe4-11ee-8f68-b7b583e7c05a	0	1	0	0	0	0	0	0	0	...	NaN	1	7	1	1	
2398	acuity-AKP-72Cxz1721419664aShUu	0	1	0	0	0	0	0	0	0	...	NaN	1	2	2	4	
2858	jtn-za370CyvD9fcoZn	1	0	0	0	0	0	0	0	0	...	NaN	6	5	6	5	
3084	jtn-hkhB0CD7IlkaL75	1	0	0	0	0	0	0	0	0	...	NaN	4	2	4	4	
3085	acuity-AKP-72Cxz1721419664aShUu	0	1	0	0	0	0	0	0	0	...	NaN	1	2	2	4	
3086	talkonline-66b2a870-abe4-11ee-8f68-b7b583e7c05a	0	1	0	0	0	0	0	0	0	...	NaN	1	7	1	1	
3087	jtn-X3x1fr7Ea1cwa2o	0	1	0	0	0	0	0	0	0	...	NaN	3	3	4	4	
3088	acuity-AKP-7480ab55-aed7-457c-95ca-20fbd562d6df	0	1	0	0	0	0	0	0	0	...	NaN	3	3	3	3	
3089	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	0	1	0	0	0	0	0	0	0	...	NaN	4	4	4	4	
3090	jtn-vG9F16NqfRwxVLx	0	1	0	0	0	0	0	0	0	...	NaN	6	3	6	8	
3091	jtn-4guyjfQ0PtYI91	0	1	0	0	0	0	0	0	0	...	NaN	1	1	1	8	
3092	jtn-za370CyvD9fcoZn	1	0	0	0	0	0	0	0	0	...	NaN	6	5	6	5	
3093	jtn-uqgmsyrCPGOMYf8	1	0	0	0	0	0	0	0	0	...	NaN	4	4	4	4	

22 rows × 17 columns

df[df.duplicated()]

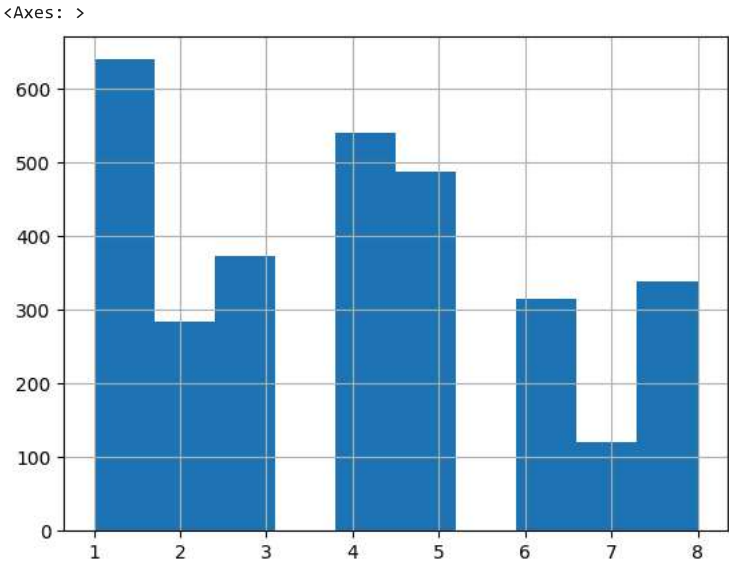
	respondent_id	q2_1	q2_2	q2_3	q3_18	q3_19	q3_20	q3_21	q3_22	q3_23	...	Q24_6_TEXT	q7_1	q7_2	q7_3	q7_4	q7_5
570	jtn-nffYkv190v2Z34A	1	0	0	0	0	0	0	0	0	...	NaN	3	3	3	3	
3084	jtn-hkhB0CD7IlkaL75	1	0	0	0	0	0	0	0	0	...	NaN	4	2	4	4	
3085	acuity-AKP-72Cxz1721419664aShUu	0	1	0	0	0	0	0	0	0	...	NaN	1	2	2	4	
3086	talkonline-66b2a870-abe4-11ee-8f68-b7b583e7c05a	0	1	0	0	0	0	0	0	0	...	NaN	1	7	1	1	
3087	jtn-X3x1fr7Ea1cwa2o	0	1	0	0	0	0	0	0	0	...	NaN	3	3	4	4	
3088	acuity-AKP-7480ab55-aed7-457c-95ca-20fbd562d6df	0	1	0	0	0	0	0	0	0	...	NaN	3	3	3	3	
3089	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	0	1	0	0	0	0	0	0	0	...	NaN	4	4	4	4	
3090	jtn-vG9F16NqfRwxVLx	0	1	0	0	0	0	0	0	0	...	NaN	6	3	6	8	
3091	jtn-4guyjfQ0PtcYl91	0	1	0	0	0	0	0	0	0	...	NaN	1	1	1	8	
3092	jtn-za370CyvD9fcoZn	1	0	0	0	0	0	0	0	0	...	NaN	6	5	6	5	
3093	jtn-uqgmsyrCPGOMYf8	1	0	0	0	0	0	0	0	0	...	NaN	4	4	4	4	

11 rows × 17 columns

```
raw_df.duplicated().sum()
```

```
np.int64(11)
```

```
raw_df['q7_1'].hist(bins=10)
```



```
df_q7_long.groupby(['question_base', 'question_id']).agg('count')
```

		respondent_id	precode	description
question_base	question_id			
q7_1	q7_1_1	639	639	639
	q7_1_2	283	283	283
	q7_1_3	372	372	372
	q7_1_4	539	539	539
	q7_1_5	487	487	487
	q7_1_6	315	315	315
	q7_1_7	121	121	121
	q7_1_8	338	338	338
q7_2	q7_2_1	514	514	514
	q7_2_2	211	211	211
	q7_2_3	387	387	387
	q7_2_4	641	641	641
	q7_2_5	494	494	494
	q7_2_6	270	270	270
	q7_2_7	157	157	157
	q7_2_8	420	420	420
q7_3	q7_3_1	490	490	490
	q7_3_2	181	181	181
	q7_3_3	323	323	323
	q7_3_4	743	743	743
	q7_3_5	446	446	446
	q7_3_6	265	265	265
	q7_3_7	160	160	160
	q7_3_8	486	486	486
q7_4	q7_4_1	517	517	517
	q7_4_2	158	158	158
	q7_4_3	248	248	248
	q7_4_4	1334	1334	1334
	q7_4_5	245	245	245
	q7_4_6	144	144	144
	q7_4_7	96	96	96
	q7_4_8	352	352	352
q7_5	q7_5_1	645	645	645
	q7_5_2	173	173	173
	q7_5_3	381	381	381
	q7_5_4	791	791	791
	q7_5_5	437	437	437
	q7_5_6	239	239	239
	q7_5_7	124	124	124
	q7_5_8	304	304	304

```
df_q7_long.groupby(['question_id','question_base']).agg('count')
```

		respondent_id	precode	description
question_id	question_base			
q7_1_1	q7_1	639	639	639
q7_1_2	q7_1	283	283	283
q7_1_3	q7_1	372	372	372
q7_1_4	q7_1	539	539	539
q7_1_5	q7_1	487	487	487
q7_1_6	q7_1	315	315	315
q7_1_7	q7_1	121	121	121
q7_1_8	q7_1	338	338	338
q7_2_1	q7_2	514	514	514
q7_2_2	q7_2	211	211	211
q7_2_3	q7_2	387	387	387
q7_2_4	q7_2	641	641	641
q7_2_5	q7_2	494	494	494
q7_2_6	q7_2	270	270	270
q7_2_7	q7_2	157	157	157
q7_2_8	q7_2	420	420	420
q7_3_1	q7_3	490	490	490
q7_3_2	q7_3	181	181	181
q7_3_3	q7_3	323	323	323
q7_3_4	q7_3	743	743	743
q7_3_5	q7_3	446	446	446
q7_3_6	q7_3	265	265	265
q7_3_7	q7_3	160	160	160
q7_3_8	q7_3	486	486	486
q7_4_1	q7_4	517	517	517
q7_4_2	q7_4	158	158	158
q7_4_3	q7_4	248	248	248
q7_4_4	q7_4	1334	1334	1334
q7_4_5	q7_4	245	245	245
q7_4_6	q7_4	144	144	144
q7_4_7	q7_4	96	96	96
q7_4_8	q7_4	352	352	352
q7_5_1	q7_5	645	645	645
q7_5_2	q7_5	173	173	173
q7_5_3	q7_5	381	381	381
q7_5_4	q7_5	791	791	791
q7_5_5	q7_5	437	437	437
q7_5_6	q7_5	239	239	239
q7_5_7	q7_5	124	124	124
q7_5_8	q7_5	304	304	304

```
df.groupby(['gender', 'q7_1']).size().reset_index(name='count')
```

	gender	q7_1	count
0	Female	1	303
1	Female	2	194
2	Female	3	206
3	Female	4	252
4	Female	5	191
5	Female	6	161
6	Female	7	61
7	Female	8	177
8	Male	1	336
9	Male	2	89
10	Male	3	166
11	Male	4	287
12	Male	5	296
13	Male	6	154
14	Male	7	60
15	Male	8	161

```
raw_df['total_selected_q7'].value_counts()
```

	count
total_selected_q7	
5	2373
4	285
0	201
3	133
2	64
1	38

```
dtype: int64
```

```
raw_df['total_selected_q7'] = raw_df[['q7_1', 'q7_2', 'q7_3', 'q7_4', 'q7_5']].apply(lambda row: sum(row != 8), axis=1)
raw_df
```

	respondent_id	q2_1	q2_2	q2_3	q3_18	q3_19	q3_20	q3_21	q3_22	q3_23	...	Q24_6_TEXT	q7_1	q7_2	q7_3	q7_4	q7_5	category
0	jtn-31kgg8psdBCqYD6	1	0	0	0	0	0	0	0	0	...	NaN	8	8	8	8	8	R
1	jtn-5FQ679JNrH6g7dT	1	0	0	0	0	0	0	0	0	...	NaN	4	1	3	1	4	R
2	jtn-22853lleE3oxpXj	1	0	0	0	0	0	0	0	0	...	NaN	4	4	4	4	4	H
3	jtn-1491R1Uo7f7tb0f	0	1	0	0	0	0	0	0	0	...	NaN	8	8	8	8	8	R
4	jtn-2UaYA52mtFm5pF3	1	0	0	0	0	0	0	0	0	...	NaN	5	5	5	5	6	H
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
3089	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	0	1	0	0	0	0	0	0	0	...	NaN	4	4	4	4	4	
3090	jtn-vG9F16NqfRwxVLx	0	1	0	0	0	0	0	0	0	...	NaN	6	3	6	8	8	E
3091	jtn-4guyjfQ0PtcYl91	0	1	0	0	0	0	0	0	0	...	NaN	1	1	1	8	1	R
3092	jtn-za370CyvD9fcoZn	1	0	0	0	0	0	0	0	0	...	NaN	6	5	6	5	6	H
3093	jtn-uqgmsyrCPGOMYf8	1	0	0	0	0	0	0	0	0	...	NaN	4	4	4	4	1	R

3094 rows × 71 columns

```
raw_df['gender'] = np.select(
    [raw_df['q2_1'] == 1, raw_df['q2_2'] == 1],
    ['Male', 'Female'],
    default='Unknown'
)
raw_df
```

	respondent_id	q2_1	q2_2	q2_3	q3_18	q3_19	q3_20	q3_21	q3_22	q3_23	...	Q22_1_TEXT	Q24_6_TEXT	q7_1	q7_2	q7_3	q7_4	q7_5	category
0	jtn-31kgg8psdBCqYD6	1	0	0	0	0	0	0	0	0	...	NaN	NaN	8	8	8	8	8	
1	jtn-5FQ679JNrH6g7dT	1	0	0	0	0	0	0	0	0	...	NaN	NaN	4	1	3	1	4	
2	jtn-22853lleE3oxpXj	1	0	0	0	0	0	0	0	0	...	NaN	NaN	4	4	4	4	4	
3	jtn-1491R1Uo7f7tb0f	0	1	0	0	0	0	0	0	0	...	NaN	NaN	8	8	8	8	8	
4	jtn-2UaYA52mtFm5pF3	1	0	0	0	0	0	0	0	0	...	NaN	NaN	5	5	5	5	6	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
3089	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	0	1	0	0	0	0	0	0	0	...	NaN	NaN	4	4	4	4	4	
3090	jtn-vG9F16NqfRwxVLx	0	1	0	0	0	0	0	0	0	...	NaN	NaN	6	3	6	8	8	
3091	jtn-4guyjfQ0PtcYl91	0	1	0	0	0	0	0	0	0	...	NaN	NaN	1	1	1	8	1	
3092	jtn-za370CyvD9fcoZn	1	0	0	0	0	0	0	0	0	...	NaN	NaN	6	5	6	5	6	
3093	jtn-uqgmsyrCPGOMYf8	1	0	0	0	0	0	0	0	0	...	NaN	NaN	4	4	4	4	1	

3094 rows × 70 columns

```
raw_df[(raw_df['q2_1'] == 1) & (raw_df['q7_1'] != 8)].shape

(1388, 69)
```



```
# Filter with regex
gender_cols = raw_df.filter(regex='^q2_').columns
df[gender_cols]
```

	q2_1	q2_2	q2_3
0	1	0	0
1	1	0	0
2	1	0	0
3	0	1	0
4	1	0	0
...	...	...	...
3089	0	1	0
3090	0	1	0
3091	0	1	0
3092	1	0	0
3093	1	0	0

3094 rows × 3 columns

```
raw_df.filter(like='q7_')
```

	q7_1	q7_2	q7_3	q7_4	q7_5
0	8	8	8	8	8
1	4	1	3	1	4
2	4	4	4	4	4
3	8	8	8	8	8
4	5	5	5	5	6
...	...	...	...	...	...
3089	4	4	4	4	4
3090	6	3	6	8	8
3091	1	1	1	8	1
3092	6	5	6	5	6
3093	4	4	4	4	1

3094 rows × 5 columns

```
raw_df.columns.tolist()
```

```
'q3_25',
'q3_26',
'q3_27',
'q3_28',
'q3_29',
'q3_30',
'q3_31',
'q3_32',
'q3_33',
'q3_34',
'q3_35',
'q3_36',
'q3_37',
```

```
q3_49',
'q3_50',
'q3_51',
'q3_52',
'q3_53',
'q3_54',
'q3_55',
'q3_56',
'q3_57',
'q3_58',
'q3_59',
'q3_60',
'q3_61',
'q3_62',
'q3_63',
'q3_64',
'q48_1',
'q48_2',
'q48_3',
's2_359',
's2_36',
's2_40',
's2_420',
's2_48',
'Q19_1_TEXT',
'Q22_1_TEXT',
'Q24_6_TEXT',
'q7_1',
'q7_2',
'q7_3',
'q7_4',
'q7_5',
'country',
'area']
```

```
raw_df['q7_1'].value_counts(dropna=False)
```

count	
q7_1	
1	639
4	539
5	487
3	372
8	338
6	315
2	283
7	121

dtype: int64

```
raw_df.isnull().sum().sort_values(ascending=False)
```

```
0
Q24_6_TEXT 3090
Q22_1_TEXT 3077
Q19_1_TEXT 3063
q2_2 0
q2_1 0
... ..
q7_1 0
q7_2 0
q7_3 0
q7_4 0
q7_5 0
```

67 rows × 1 columns

dtype: int64

```
raw_df[raw_df.isnull().any(axis=1)]
```

	respondent_id	q2_1	q2_2	q2_3	q3_18	q3_19	q3_20	q3_21	q3_22	q3_23	...	s2_420	s2_48	Q19_1_TEXT	Q22_1_TEXT	Q24_1
0	jtn-31kgg8psdBCqYD6	1	0	0	0	0	0	0	0	0	...	1	0	NaN	NaN	
1	jtn-5FQ679JNrH6g7dT	1	0	0	0	0	0	0	0	0	...	1	0	NaN	NaN	
2	jtn-22853lleE3oxpXj	1	0	0	0	0	0	0	0	0	...	0	0	NaN	NaN	
3	jtn-1491R1Uo7f7tb0f	0	1	0	0	0	0	0	0	0	...	1	0	NaN	NaN	
4	jtn-2UaYA52mtFm5pF3	1	0	0	0	0	0	0	0	0	...	0	0	NaN	NaN	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
3089	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	0	1	0	0	0	0	0	0	0	...	0	1	NaN	NaN	
3090	jtn-vG9F16NqfRwxVLx	0	1	0	0	0	0	0	0	0	...	0	0	NaN	NaN	
3091	jtn-4guyjfQ0PtcYI91	0	1	0	0	0	0	0	0	0	...	1	0	NaN	NaN	
3092	jtn-za370CyvD9fcoZn	1	0	0	0	0	0	0	0	0	...	0	0	NaN	NaN	
3093	jtn-uqgmsyrCPGOMYf8	1	0	0	0	0	0	0	0	0	...	1	0	NaN	NaN	

3094 rows × 67 columns

```
labels_dict = labels_df.set_index('Question ID')['Base option'].to_dict()
labels_dict
```

```
q48_2 : Suburban ,
'q48_3': 'Rural',
'q7_1_1': 'Help me discover a new brand, product or service',
'q7_1_2': 'Help me discover a new brand, product or service',
'q7_1_3': 'Help me discover a new brand, product or service',
'q7_1_4': 'Help me discover a new brand, product or service',
'q7_1_5': 'Help me discover a new brand, product or service',
'q7_1_6': 'Help me discover a new brand, product or service',
'q7_1_7': 'Help me discover a new brand, product or service',
'q7_1_8': 'Help me discover a new brand, product or service',
'q7_2_1': 'Help me determine the quality',
'q7_2_2': 'Help me determine the quality',
'q7_2_3': 'Help me determine the quality',
'q7_2_4': 'Help me determine the quality',
'q7_2_5': 'Help me determine the quality',
'q7_2_6': 'Help me determine the quality',
'q7_2_7': 'Help me determine the quality',
'q7_2_8': 'Help me determine the quality',
'q7_3_1': 'To give my confidence in my choice',
'q7_3_2': 'To give my confidence in my choice',
'q7_3_3': 'To give my confidence in my choice',
'q7_3_4': 'To give my confidence in my choice',
'q7_3_5': 'To give my confidence in my choice',
'q7_3_6': 'To give my confidence in my choice',
'q7_3_7': 'To give my confidence in my choice',
'q7_3_8': 'To give my confidence in my choice',
'q7_4_1': "To reassure me it's the right price",
'q7_4_2': "To reassure me it's the right price",
'q7_4_3': "To reassure me it's the right price",
'q7_4_4': "To reassure me it's the right price",
'q7_4_5': "To reassure me it's the right price",
'q7_4_6': "To reassure me it's the right price",
'q7_4_7': "To reassure me it's the right price",
'q7_4_8': "To reassure me it's the right price",
'q7_5_1': 'To help me find the best brand/product relevant to my needs',
'q7_5_2': 'To help me find the best brand/product relevant to my needs',
'q7_5_3': 'To help me find the best brand/product relevant to my needs',
'q7_5_4': 'To help me find the best brand/product relevant to my needs',
'q7_5_5': 'To help me find the best brand/product relevant to my needs',
'q7_5_6': 'To help me find the best brand/product relevant to my needs',
'q7_5_7': 'To help me find the best brand/product relevant to my needs',
'q7_5_8': 'To help me find the best brand/product relevant to my needs'}
```

```
question_map = labels_df.groupby('Question Code')[['Question']].first()
question_map
question_map[question_map.index=='q7']
```

Question	
Question Code	
q7	For each of the purposes listed below, which o...

```
question_map = labels_df.groupby('Question Code')[['Question', 'Question Code']].first()
question_map
```

Question		Question Code
Question Code		
q2	Which of the following best describes your gen...	q2
q3	How old are you?	q3
q3a	Age groups	q3a
q48	Which best describes the area where you live?	q48
q7	For each of the purposes listed below, which o...	q7
s2	What country do you live in currently?	s2

```
df= raw_df
q7_cols = [col for col in df.columns if col.startswith('q7_')]
df_q7 = df[['respondent_id'] + q7_cols]
df_q7
```

	respondent_id	q7_1	q7_2	q7_3	q7_4	q7_5
0	jtn-31kgg8psdBCqYD6	8	8	8	8	8
1	jtn-5FQ679JNrH6g7dT	4	1	3	1	4
2	jtn-22853lleE3oxpXj	4	4	4	4	4
3	jtn-1491R1Uo7f7tb0f	8	8	8	8	8
4	jtn-2UaYA52mtFm5pF3	5	5	5	5	6
...	...	...	...	...	...	...
3089	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	4	4	4	4	4
3090	jtn-vG9F16NqfRwxVLx	6	3	6	8	8
3091	jtn-4guyjfQ0PtcYl91	1	1	1	8	1
3092	jtn-za370CyvD9fcoZn	6	5	6	5	6
3093	jtn-uqgmsyrCPGOMYf8	4	4	4	4	1

3094 rows × 6 columns

```
df_q7_long = df_q7.melt(id_vars='respondent_id',
                        var_name='question_id',
                        value_name='selected')
```

df_q7_long

	respondent_id	question_id	selected
0	jtn-31kgg8psdBCqYD6	q7_1	8
1	jtn-5FQ679JNrH6g7dT	q7_1	4
2	jtn-22853lleE3oxpXj	q7_1	4
3	jtn-1491R1Uo7f7tb0f	q7_1	8
4	jtn-2UaYA52mtFm5pF3	q7_1	5
...	...	...	...
15465	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	q7_5	4
15466	jtn-vG9F16NqfRwxVLx	q7_5	8
15467	jtn-4guyjfQ0PtcYl91	q7_5	1
15468	jtn-za370CyvD9fcoZn	q7_5	6
15469	jtn-uqgmsyrCPGOMYf8	q7_5	1

15470 rows × 3 columns

```
df_q7_long[df_q7_long['selected'] == 1].shape[0] / df_q7_long.shape[0]
```

0.1813186813186813

```
df_q7_long[df_q7_long['selected'] == 1]
```

	respondent_id	question_id	selected
9	jtn-92P5tz6B5EzG84R	q7_1	1
10	jtn-08a82z0U2p1KBic	q7_1	1
13	jtn-2xs55WAh4e120r9	q7_1	1
14	jtn-0Dq29RYJ08Wpq1h	q7_1	1
20	jtn-H8W4s940wNAJOb3	q7_1	1
...	...	...	...
15420	jtn-T957g7DJBUEIKNJ-gwi-vyr-gke	q7_5	1
15426	jtn-3lqj805vD3gZIJ7-gwi-vyr-gke	q7_5	1
15462	talkonline-66b2a870-abe4-11ee-8f68-b7b583e7c05a	q7_5	1
15467	jtn-4guyjfQ0PtcYI91	q7_5	1
15469	jtn-uqgmsyrCPGOMYf8	q7_5	1

2805 rows × 3 columns

```
df_q7_long['description'] = df_q7_long['question_id'].map(labels_dict)
df_q7_long
```

	respondent_id	question_id	selected	description
0	jtn-31kgg8psdBCqYD6	q7_1	8	NaN
1	jtn-5FQ679JNrH6g7dT	q7_1	4	NaN
2	jtn-22853lleE3oxpXj	q7_1	4	NaN
3	jtn-1491R1Uo7f7tb0f	q7_1	8	NaN
4	jtn-2UaYA52mtFm5pF3	q7_1	5	NaN
...	...	...	...	...
15465	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	q7_5	4	NaN
15466	jtn-vG9F16NqfRwxVLx	q7_5	8	NaN
15467	jtn-4guyjfQ0PtcYI91	q7_5	1	NaN
15468	jtn-za370CyyD9fcoZn	q7_5	6	NaN
15469	jtn-uqgmsyrCPGOMYf8	q7_5	1	NaN

15470 rows × 4 columns

```
gender_map = {'q2_1': 'Male', 'q2_2': 'Female'}
df_gender = df[['respondent_id', 'q2_1', 'q2_2']].melt('respondent_id')
df_gender
```

	respondent_id	variable	value
0	jtn-31kgg8psdBCqYD6	q2_1	1
1	jtn-5FQ679JNrH6g7dT	q2_1	1
2	jtn-22853lleE3oxpXj	q2_1	1
3	jtn-1491R1Uo7f7tb0f	q2_1	0
4	jtn-2UaYA52mtFm5pF3	q2_1	1
...	...	...	...
6183	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	q2_2	1
6184	jtn-vG9F16NqfRwxVLx	q2_2	1
6185	jtn-4guyjfQ0PtcYI91	q2_2	1
6186	jtn-za370CyyD9fcoZn	q2_2	0
6187	jtn-uqgmsyrCPGOMYf8	q2_2	0

6188 rows × 3 columns

```
gender_map = {'q2_1': 'Male', 'q2_2': 'Female'}
df_gender = df[['respondent_id', 'q2_1', 'q2_2']].melt('respondent_id')
df_gender = df_gender[df_gender['value'] == 1]
df_gender['gender'] = df_gender['variable'].map(gender_map)
df_gender
```

	respondent_id	variable	value	gender
0	jtn-31kgg8psdBCqYD6	q2_1	1	Male
1	jtn-5FQ679JNrH6g7dT	q2_1	1	Male
2	jtn-22853lleE3oxpXj	q2_1	1	Male
4	jtn-2UaYA52mtFm5pF3	q2_1	1	Male
6	jtn-4Nu4uiWsxbwLn4r	q2_1	1	Male
...	...	...	...	...
6181	jtn-X3x1fr7Ea1cwa2o	q2_2	1	Female
6182	acuity-AKP-7480ab55-aed7-457c-95ca-20fbd562d6df	q2_2	1	Female
6183	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	q2_2	1	Female
6184	jtn-vG9F16NqfRwxVLx	q2_2	1	Female
6185	jtn-4guyjfQ0PtcYl91	q2_2	1	Female

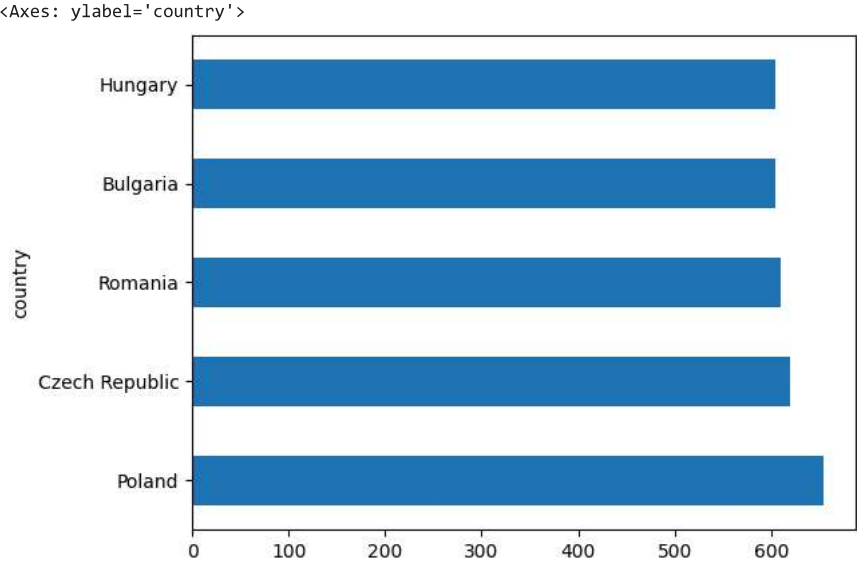
3094 rows × 4 columns

```
country_cols = [col for col in df.columns if col.startswith('s2_')]
df['country'] = df[country_cols].idxmax(axis=1).map(labels_dict)
df['country'].value_counts()
```

	count
country	
Poland	655
Czech Republic	620
Romania	610
Bulgaria	605
Hungary	604

dtype: int64

```
country_cols = [col for col in df.columns if col.startswith('s2_')]
df['country'] = df[country_cols].idxmax(axis=1).map(labels_dict)
df['country'].value_counts().plot(kind='barh')
```



df[q7_cols]

	q7_1	q7_2	q7_3	q7_4	q7_5
0	8	8	8	8	8
1	4	1	3	1	4
2	4	4	4	4	4
3	8	8	8	8	8
4	5	5	5	5	6
...	...	...	...	...	...
3089	4	4	4	4	4
3090	6	3	6	8	8
3091	1	1	1	8	1
3092	6	5	6	5	6
3093	4	4	4	4	1

3094 rows × 5 columns

df[q7_cols].values

```
array([[8, 8, 8, 8, 8],
       [4, 1, 3, 1, 4],
       [4, 4, 4, 4, 4],
       ...,
       [1, 1, 1, 8, 1],
       [6, 5, 6, 5, 6],
       [4, 4, 4, 4, 1]])
```

df[q7_cols].values.shape

(3094, 5)

df[q7_cols].values.reshape(5, -1)

```
array([[8, 8, 8, ..., 1, 2, 5],
       [4, 8, 8, ..., 7, 4, 8],
       [4, 6, 5, ..., 4, 7, 7],
       [7, 7, 7, ..., 7, 7, 3],
       [3, 3, 3, ..., 4, 4, 1]])
```

df[q7_cols].values.reshape(5, -1).shape

(5, 3094)

```
q7_matrix = df[q7_cols].replace({8: np.nan})
q7_matrix
```

	q7_1	q7_2	q7_3	q7_4	q7_5
0	NaN	NaN	NaN	NaN	NaN
1	4.0	1.0	3.0	1.0	4.0
2	4.0	4.0	4.0	4.0	4.0
3	NaN	NaN	NaN	NaN	NaN
4	5.0	5.0	5.0	5.0	6.0
...	...	...	...	...	...
3089	4.0	4.0	4.0	4.0	4.0
3090	6.0	3.0	6.0	NaN	NaN
3091	1.0	1.0	1.0	NaN	1.0
3092	6.0	5.0	6.0	5.0	6.0
3093	4.0	4.0	4.0	4.0	1.0

3094 rows × 5 columns


```
q7_matrix.isna()
q7_matrix.isna().mean()
```

```

      0
q7_1  0.109244
q7_2  0.135747
q7_3  0.157078
q7_4  0.113769
q7_5  0.098255

dtype: float64
```

```
q7_matrix.isna().mean().values
```

```
array([0.1092437 , 0.13574661, 0.15707822, 0.11376858, 0.09825469])
```

```
q7_matrix.isna().mean().values.reshape(5, -1).shape
```

```
(5, 1)
```

```
q7_matrix.isna().mean().values.reshape(5, -1)
```

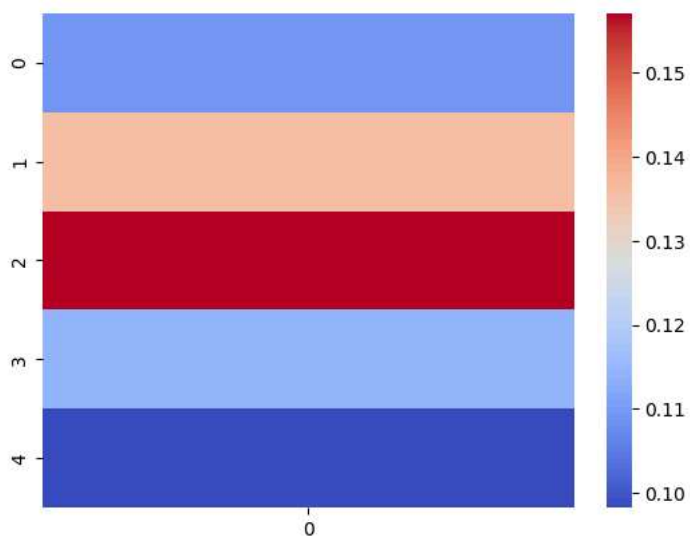
```
array([[0.1092437 ],
       [0.13574661],
       [0.15707822],
       [0.11376858],
       [0.09825469]])
```

```
q7_matrix.isna().mean().values.reshape(5, -1).shape
```

```
(5, 1)
```

```
import seaborn as sns
q7_matrix = df[q7_cols].replace({8: np.nan}) # replace 'None' with NaN
sns.heatmap(q7_matrix.isna().mean().values.reshape(5, -1), cmap="coolwarm")
```

```
<Axes: >
```



```
country_cols = [col for col in df.columns if col.startswith('s2_')]
country_cols
```

```
['s2_359', 's2_36', 's2_40', 's2_420', 's2_48']
```

```
df[country_cols]
```

	s2_359	s2_36	s2_40	s2_420	s2_48
0	0	0	0	1	0
1	0	0	0	1	0
2	0	1	0	0	0
3	0	0	0	1	0
4	0	1	0	0	0
...	...	...	...	...	...
3089	0	0	0	0	1
3090	1	0	0	0	0
3091	0	0	0	1	0
3092	0	1	0	0	0
3093	0	0	0	1	0

3094 rows × 5 columns

df[country_cols][109:110]

	s2_359	s2_36	s2_40	s2_420	s2_48
109	0	0	0	0	1

df[country_cols].loc[109]

	109
s2_359	0
s2_36	0
s2_40	0
s2_420	0
s2_48	1

dtype: int64

df[country_cols].idxmax()

	0
s2_359	5
s2_36	2
s2_40	9
s2_420	0
s2_48	109

dtype: int64

df[country_cols].idxmax(axis=1)

```
0
0  s2_420
1  s2_420
2   s2_36
3  s2_420
4   s2_36
...
3089 s2_48
3090 s2_359
3091 s2_420
3092 s2_36
3093 s2_420
```

3094 rows × 1 columns

dtype: object

```
country_cols = [col for col in df.columns if col.startswith('s2_')]
df['country'] = df[country_cols].idxmax(axis=1).map(labels_dict)
df
```

	respondent_id	q2_1	q2_2	q2_3	q3_18	q3_19	q3_20	q3_21	q3_22	q3_23	...	s2_48	Q19_1_TEXT	Q22_1_TEXT	Q24_6_TEXT
0	jtn-31kgg8psdBCqYD6	1	0	0	0	0	0	0	0	0	...	0	NaN	NaN	NaN
1	jtn-5FQ679JNrH6g7dT	1	0	0	0	0	0	0	0	0	...	0	NaN	NaN	NaN
2	jtn-22853lleE3oxpXj	1	0	0	0	0	0	0	0	0	...	0	NaN	NaN	NaN
3	jtn-1491R1Uo7f7tb0f	0	1	0	0	0	0	0	0	0	...	0	NaN	NaN	NaN
4	jtn-2UaYA52mtFm5pF3	1	0	0	0	0	0	0	0	0	...	0	NaN	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
3089	toluna-5729c24e-851e-4f6f-91e6-e7fdf8c6dc5f	0	1	0	0	0	0	0	0	0	...	1	NaN	NaN	NaN
3090	jtn-vG9F16NqfRwxVLx	0	1	0	0	0	0	0	0	0	...	0	NaN	NaN	NaN
3091	jtn-4guyjfQ0PtcYI91	0	1	0	0	0	0	0	0	0	...	0	NaN	NaN	NaN
3092	jtn-za370CyyD9fcoZn	1	0	0	0	0	0	0	0	0	...	0	NaN	NaN	NaN
3093	jtn-uqgmsyrCPGOMYf8	1	0	0	0	0	0	0	0	0	...	0	NaN	NaN	NaN

3094 rows × 68 columns

```
df['country'].value_counts()
```

	count
country	
Poland	655
Czech Republic	620
Romania	610
Bulgaria	605
Hungary	604

dtype: int64

```
q7_cols = [col for col in df.columns if col.startswith('q7_')]
df_q7_long = df[['respondent_id'] + q7_cols].melt('respondent_id')
df_q7_long = df_q7_long[df_q7_long['value'] != 0]
df_q7_long['label'] = df_q7_long['variable'].map(labels_dict)
df_q7_long
```

	respondent_id	variable	value	label
0	jtn-31kgg8psdBCqYD6	q7_1	8	NaN
1	jtn-5FQ679JNrH6g7dT	q7_1	4	NaN
2	jtn-22853lleE3oxpXj	q7_1	4	NaN
3	jtn-1491R1Uo7f7tb0f	q7_1	8	NaN
4	jtn-2UaYA52mtFm5pF3	q7_1	5	NaN
...	...	...	...	...
15465	toluna-5729c24e-851e-4f6f-91e6-e7df8c6dc5f	q7_5	4	NaN
15466	jtn-vG9F16NqfRwxVLx	q7_5	8	NaN
15467	jtn-4guyjfQ0PtcYl91	q7_5	1	NaN
15468	jtn-za370CyvD9fcoZn	q7_5	6	NaN
15469	jtn-uqgmsyrCPGOMYf8	q7_5	1	NaN

15470 rows × 4 columns

```
set(labels_df['Question ID']).intersection(set(raw_df.columns))
```

```
{'q2_1',
 'q2_2',
 'q3_18',
 'q3_19',
 'q3_20',
 'q3_21',
 'q3_22',
 'q3_23',
 'q3_24',
 'q3_25',
 'q3_26',
 'q3_27',
 'q3_28',
 'q3_29',
 'q3_30',
 'q3_31',
 'q3_32',
 'q3_33',
 'q3_34',
 'q3_35',
 'q3_36',
 'q3_37',
 'q3_38',
 'q3_39',
 'q3_40',
 'q3_41',
 'q3_42',
 'q3_43',
 'q3_44',
 'q3_45',
 'q3_46',
 'q3_47',
 'q3_48',
```

```
'q3_49',
'q3_50',
'q3_51',
'q3_52',
'q3_53',
'q3_54',
'q3_55',
'q3_56',
'q3_57',
'q3_58',
'q3_59',
'q3_60',
'q3_61',
'q3_62',
'q3_63',
'q3_64',
'q48_1',
'q48_2',
'q48_3',
's2_359',
's2_36',
's2_40',
's2_420',
's2_48'}
```

```
missing_cols = set(labels_df['Question ID']) - set(raw_df.columns)
missing_cols
```

```
{'q3a_1',
'q3a_2',
'q3a_3',
'q3a_4',
'q3a_5',
'q3a_6',
'q7_1_1',
'q7_1_2',
'q7_1_3',
'q7_1_4',
'q7_1_5',
'q7_1_6',
'q7_1_7',
'q7_1_8',
'q7_2_1',
'q7_2_2',
'q7_2_3',
'q7_2_4',
'q7_2_5',
'q7_2_6',
'q7_2_7',
'q7_2_8',
'q7_3_1',
'q7_3_2',
'q7_3_3',
'q7_3_4',
'q7_3_5',
'q7_3_6',
'q7_3_7',
'q7_3_8',
'q7_4_1',
'q7_4_2',
'q7_4_3',
'q7_4_4',
'q7_4_5',
'q7_4_6',
'q7_4_7',
'q7_4_8',
'q7_5_1',
'q7_5_2',
'q7_5_3',
'q7_5_4',
'q7_5_5',
'q7_5_6',
'q7_5_7',
'q7_5_8'}
```

```
extra_cols = set(raw_df.columns) - set(labels_df['Question ID']) - {'respondent_id'}
extra_cols
```

```
{'Q19_1_TEXT',
'Q22_1_TEXT',
'Q24_6_TEXT',
'country',
```

```
'q2_3',
'q7_1',
'q7_2',
'q7_3',
'q7_4',
'q7_5'}
```

```
raw_df[labels_df['Question ID'].unique()].apply(pd.Series.value_counts).T.fillna(0)
```

```
-----
KeyError                                Traceback (most recent call last)
/tmp/ipython-input-75-3474193641.py in <cell line: 0>()
----> 1 raw_df[labels_df['Question ID'].unique()].apply(pd.Series.value_counts).T.fillna(0)

----- 2 frames -----
/usr/local/lib/python3.11/dist-packages/pandas/core/indexes/base.py in _raise_if_missing(self, key, indexer, axis_name)
    6250
    6251         not_found = list(ensure_index(key)[missing_mask.nonzero()[0]].unique())
-> 6252         raise KeyError(f"{not_found} not in index")
    6253
    6254     @overload

KeyError: "[ 'q3a_1', 'q3a_2', 'q3a_3', 'q3a_4', 'q3a_5', 'q3a_6', 'q7_1_1', 'q7_1_2', 'q7_1_3', 'q7_1_4', 'q7_1_5', 'q7_1_6',
'q7_1_7', 'q7_1_8', 'q7_2_1', 'q7_2_2', 'q7_2_3', 'q7_2_4', 'q7_2_5', 'q7_2_6', 'q7_2_7', 'q7_2_8', 'q7_3_1', 'q7_3_2', 'q7_3_3',
'q7_3_4', 'q7_3_5', 'q7_3_6', 'q7_3_7', 'q7_3_8', 'q7_4_1', 'q7_4_2', 'q7_4_3', 'q7_4_4', 'q7_4_5', 'q7_4_6', 'q7_4_7', 'q7_4_8',
'q7_5_1', 'q7_5_2', 'q7_5_3', 'q7_5_4', 'q7_5_5', 'q7_5_6', 'q7_5_7', 'q7_5_8'] not in index"
```

```
raw_df['q7_5'].value_counts()
```

count	
q7_5	
4	791
1	645
5	437
3	381
8	304
6	239
2	173
7	124

dtype: int64

```
raw_df['q7_1'].value_counts(dropna=False)
```

count	
q7_1	
1	639
4	539
5	487
3	372
8	338
6	315
2	283
7	121

dtype: int64

```
def extract_country(row):
    for col in row.index:
        if col.startswith('s2_') and row[col] == 1:
            return labels_df.loc[labels_df['Question ID'] == col, 'Base option'].values[0]
    return None
```

```
raw_df['country'] = raw_df.apply(extract_country, axis=1)
raw_df['country']
```

	country
0	Czech Republic
1	Czech Republic
2	Hungary
3	Czech Republic
4	Hungary
...	...
3089	Poland
3090	Bulgaria
3091	Czech Republic
3092	Hungary
3093	Czech Republic

3094 rows × 1 columns

dtype: object

```
def extract_country(row):
    for col in row.index:
        if col.startswith('s2_') and row[col] == 1:
            return labels_df.loc[labels_df['Question ID'] == col, 'Base option'].values[0]
    return None
raw_df['country'] = raw_df.apply(extract_country, axis=1)
raw_df['country'].value_counts()
```

	count
country	
Poland	655
Czech Republic	620
Romania	610
Bulgaria	605
Hungary	604

dtype: int64

```
raw_df[['q2_1', 'q2_2']]
```

	q2_1	q2_2
0	1	0
1	1	0
2	1	0
3	0	1
4	1	0
...	...	...
3089	0	1
3090	0	1
3091	0	1
3092	1	0
3093	1	0

3094 rows × 2 columns

```
raw_df[['q2_1', 'q2_2']].idxmax(axis=1)
```

	0
0	q2_1
1	q2_1
2	q2_1
3	q2_2
4	q2_1
...	...
3089	q2_2
3090	q2_2
3091	q2_2
3092	q2_1
3093	q2_1

3094 rows × 1 columns

dtype: object

```
raw_df[['q2_1', 'q2_2']].idxmax(axis=1).str[-1]
```

	0
0	1
1	1
2	1
3	2
4	1
...	...
3089	2
3090	2
3091	2
3092	1
3093	1

3094 rows × 1 columns

dtype: object

```
raw_df[['q2_1', 'q2_2']].idxmax(axis=1).str[-1].map({'1': 'Male', '2': 'Female'})
```


0	
0	Male
1	Male
2	Male
3	Female
4	Male
...	...
3089	Female
3090	Female
3091	Female
3092	Male
3093	Male

3094 rows × 1 columns

dtype: object

```
raw_df['gender'] = raw_df[['q2_1', 'q2_2']].idxmax(axis=1).str[-1].map({'1': 'Male', '2': 'Female'})
raw_df['gender'].value_counts()
```

count	
gender	
Male	1549
Female	1545

dtype: int64

```
age_cols = [col for col in raw_df.columns if col.startswith('q3_')]
raw_df[age_cols].sum().sort_index() # Should match 18-64
```